Tool Harness Adaptation under Environment Change

PE 6203 Group Project 4 | Course delivery review copy | 31 October 2026

Research questions

Does updating affected tool knowledge recover task performance after persistent tool-environment change? Does it preserve capabilities on tasks whose tools remain unchanged? We evaluate quality, efficiency, cross-version stability, update cost and execution status separately; no weighted score is introduced.

Abstract

We implement a deterministic Selective Patch that compares tool metadata, expands the affected set through explicit dependencies, updates affected records and preserves unaffected knowledge. The same fixed DeepSeek model executes a documented three-task MCPEvol-Bench subset in Early, Middle and Late environments under No Update, Full Rewrite and Selective Patch. One execution per cell gives 27 trajectories, each with five valid judge ratings.

Independent recalculation from the frozen raw archive confirms that Selective Patch has the highest observed mean Planning Effectiveness (9.1111), but lower mean Task Fulfillment (9.8889), population ECS (9.7318) and normal completion (6/9) than both baselines. It uses zero model tokens for updating, tied with No Update, and 1.145 seconds of local update time. Its changed-task TF gain relative to No Update is 0 in Middle and +0.1 in Late; unchanged-task TF is retained.

The results establish a runnable, auditable course experiment. They do not establish general superiority, statistical significance or broad generalization. Early arms share identical knowledge, yet exhibit different truncation and completion outcomes. This variation materially affects ECS.

Submission status

The quantitative experiment and reproducibility materials are ready for review. The strict failure-case requirement remains open. Existing evidence confirms stale No Update failures, but not a natural original-Selective-Patch under-update. Two over-update candidates remain suggestive, and six description-rollback diagnostics did not establish causal harm. These limitations are reported rather than relabelled as successful completion.

No new model execution, strategy development or deployment was performed for this delivery. Historical scores, failed trajectories and billing uncertainties are retained. Any proposed diagnostic requires separate explicit approval.

Evidence: evidence/main-raw.tar.gz; evidence/recomputed/metrics.json; evidence/failure-cases.json; original course brief §§3-4.

1. Method and architecture

A harness stores tool names, input schemas, descriptions, usage rules, output schemas and routing. Persistent environment changes can make these records stale even if the base model remains fixed. Our intervention is the tool-knowledge snapshot presented to the agent, not model fine-tuning.

Adaptation pipeline

```
Early knowledge K0 + target-stage metadata
-> deterministic add / remove / field diff
-> affected records + explicit one-hop dependencies
-> copy authoritative target fields for affected records
-> preserve unaffected records and valid usage rules
-> schema / routing / preservation validation
-> frozen knowledge Kt -> common agent -> real MCP tools
-> raw trajectory + five judge ratings + independent analysis
```

Tool identity is server_id::name. The comparison covers name, parameters, description, output_schema and routing; a rename without stable identity evidence is treated as removal plus addition. Structural changes propagate through explicit tool references in usage rules. Target metadata is authoritative for structural fields. Usage rules are preserved rather than erased by an empty metadata field; unresolved semantic dependencies are recorded. The evaluated Selective Patch performs no LLM update calls.

Method	Knowledge policy
No Update	Retain the Early snapshot K0 after environmental change.
Full Rewrite	Rebuild all target records and use the fixed LLM to regenerate every target description; protect schemas, routes and supplied rules.
Selective Patch	Update affected records by deterministic diff and dependency expansion; preserve unaffected records.

All three methods use the same execution agent to choose tools, arguments and next steps. Candidate MCP servers come from the official task. No task-specific call sequence is scripted. Early uses common K0 in every arm. This implementation compares both update scope and description-generation mechanism; it is not a pure scope-only ablation.

Evidence: Frozen source src/pe6203/selective.py, baselines.py, runner.py and integration/baselines.py; per-cell adaptation/adaptation.json and knowledge.json.

2. Experimental setup and controls

Task	Role	Represented changes
133	Changed: mathematical syntax and security guidance	Middle: optional parameters; Late: required allow_variables and relevant optional tool addition.
283	Changed: analytics project decomposition	Tool additions plus parameter and description changes in planning/project tools.
235	Unchanged: supplied sales visualization	28 chart-tool records canonically unchanged; used for retention.

The fixed subset is identical across methods. TOOL coverage is limited to relevant optional additions, not mandatory renaming/removal. PARAM and DESC changes affect required tools. Metadata coverage does not mean every trajectory actually called every changed tool. Task 283 shares a service family with development task 139, so this is not an unseen-family evaluation.

Control	Frozen value
Model / provider	deepseek/deepseek-v4.1-flash-1017 via OpenRouter; same agent, judge and updater model
Decoding	temperature 0; provider seed not sent
Agent budget	4096 output tokens; 12 rounds; 900-second run budget
Timeouts / retry	request 120 s; tool 30 s; at most one HTTP retry
Updater	16384 output tokens; 120 s timeout; one HTTP retry
Judge	4096 output tokens; reasoning disabled; JSON; five first-valid slots; at most two logical attempts per slot
Measurement	Original rubric with common concise output contract and deterministic call counts appended
Execution	3 tasks x 3 stages x 3 strategies x 1 execution = 27; 135 valid rating slots

One execution is permitted by brief §3.2 under compute constraints and is explicitly disclosed. Five judge ratings are measurements of the same trajectory, not five independent agent executions. All 135 slots in this main cohort succeeded on their first logical attempt. No benchmark labels or question text were modified.

Checks from the archive confirm identical task-question hashes, model settings, budgets and static judge rubric within matched task/slot groups. Prompt evidence necessarily varies with each trajectory. Historic v1.2, v1.3 and research cohorts remain separate.

Evidence: main archive plan.json, plan.sha256, saved judge prompts, judge-attempts/*/response.json; evidence/recomputed/metrics.json fairness block.

3. Main results and execution status

Method	Mean TF	Mean PE	ECS pop.	ECS sample	Completed
No Update	9.9556	7.8667	9.9241	9.9171	7/9
Full Rewrite	9.9778	8.3556	9.9464	9.9393	7/9
Selective Patch	9.8889	9.1111	9.7318	9.6964	6/9

Each mean equally weights nine task-stage cells; each cell score averages five stored ratings. Scored trajectories with truncation or round exhaustion remain included. Selective Patch has observed PE gains of +1.2444 against No Update and +0.7556 against Full Rewrite. These are descriptive differences on one small cohort, not tested population effects.

Method / stage	TF	PE
No Update / Early	9.9333	8.6667
No Update / Middle	10.0000	6.6667
No Update / Late	9.9333	8.2667
Full Rewrite / Early	10.0000	9.0000
Full Rewrite / Middle	9.9333	7.8000
Full Rewrite / Late	10.0000	8.2667
Selective Patch / Early	9.6667	9.6667
Selective Patch / Middle	10.0000	9.1333
Selective Patch / Late	10.0000	8.5333

Method	Completed	Max rounds	Truncated
No Update	7	1	1
Full Rewrite	7	2	0
Selective Patch	6	2	1

Across all 27 trajectories: 20 completed, five exhausted their round budget and two were truncated. Normal completion means the agent loop ended normally with a non-empty answer, not independent proof of correctness. TF can be 10 even when the agent has not ended normally.

Evidence: evidence/recomputed/cells.csv; all run/result.json, raw_trace.json and immutable EVAL outputs in main archive.

4. Exact PE check and ECS conventions

Task	Stage	Five raw PE ratings	Cell PE
133	Early	10, 10, 10, 10, 10	10
133	Middle	9, 8, 8, 7, 8	8
133	Late	7, 7, 7, 7, 7	7
235	Early	10, 10, 10, 10, 10	10
235	Middle	10, 10, 10, 10, 10	10
235	Late	10, 10, 10, 10, 10	10
283	Early	9, 9, 9, 9, 9	9
283	Middle	9, 9, 9, 10, 10	9.4
283	Late	7, 9, 9, 9, 9	8.6

(10 + 8 + 7 + 10 + 10 + 10 + 9 + 9.4 + 8.6) / 9 = 82 / 9 = 9.111111... The reported 9.1111 is correct. The independent script uses exact rational arithmetic for score means; it never edits stored ratings.

Cross-version ECS

For each task, first compute the mean of its three TF values, then subtract their standard deviation. ECS is the average of these task-level quantities. It is not computed by subtracting the spread of three overall stage means.

```
mu_t = (TF_early + TF_middle + TF_late) / 3
sigma_population = sqrt(sum((TF_i - mu_t)^2) / 3)
sigma_sample = sqrt(sum((TF_i - mu_t)^2) / 2)
ECS = mean_over_tasks(mu_t - sigma_t)
```

The released implementation uses population standard deviation; the paper appendix specifies the sample convention. Both are reported, with population as the primary implementation-compatible result. For Selective Patch, task 133 TF=[9, 10, 10] gives task ECS 9.1953 (population) or 9.0893 (sample); tasks235 and 283 each give10. Therefore overall ECS is 9.7318 or9.6964.

The low Early TF on task 133 occurred with common K0 and output truncation. It reduces the Patch ECS, but cannot be attributed to a patch that was not yet applied. Conversely, higher subsequent TF is not automatically a pure update effect.

Evidence: evidence/recomputed/metrics.json: selective_PE and ecs_by_task; MCPEvol-Bench §6.1, Appendix D.1; upstream evaluation code in frozen source provenance.

5. Recovery and retention

Changed tasks: 133 and 283

Method	Stage	Mean TF	Delta vs same-stage No Update	Delta vs own Early
No Update	Middle	10.00	+0.00	+0.10
No Update	Late	9.90	+0.00	+0.00
Full Rewrite	Middle	9.90	-0.10	-0.10
Full Rewrite	Late	10.00	+0.10	+0.00
Selective Patch	Middle	10.00	+0.00	+0.50
Selective Patch	Late	10.00	+0.10	+0.50

Selective Patch matches No Update in Middle and is +0.1 TF higher in Late on the two changed tasks. Its own Early-to-Late increase is partly sensitive to the Early truncation. The more direct local mechanism is task 133 Late: four initial No Update calls omit a newly mandatory parameter and are rejected; Patch provides that parameter and has no tool errors in its corresponding run. No Update later repairs its calls using error feedback, reducing the final TF gap.

Unchanged task: 235

Method	Stage	TF	PE	Delta PE vs own Early
No Update	Middle	10.0	4.8	-5.2
No Update	Late	10.0	10.0	+0.0
Full Rewrite	Middle	10.0	10.0	+0.0
Full Rewrite	Late	10.0	10.0	+0.0
Selective Patch	Middle	10.0	10.0	+0.0
Selective Patch	Late	10.0	10.0	+0.0

Every task 235 TF is 10, so all TF retention deltas are zero. Patch retains its three chart-tool calls and PE 10. No Update Middle has five map calls and PE 4.8 despite unchanged knowledge, showing that redundant actions need not come from rewriting. Tool output consists of chart specifications: browser rendering and a hosted dashboard were not validated.

Evidence: Recomputed recovery and retention arrays; tasks133/283 labelled changed and 235 unaffected before scoring in the frozen suite.

6. Update and execution costs

Method	Update tokens	Update seconds	Update API USD
No Update	0	0.000000	0.000000000
Full Rewrite	177974	203.507975	0.049176082
Selective Patch	0	1.145192	0.000000000

Selective Patch is cheaper to update than Full Rewrite, but is tied with No Update at zero model tokens and zero API dollars. No Update also has zero measured update time, whereas Patch performs local diff and validation work. Zero API dollars does not imply zero computation or engineering cost.

Main-cohort cost component	Known USD
Agent requests	0.278473878
Judge requests	0.079262222
Adaptation	0.049176082
Preflight	0.000007110
Known total (lower bound)	0.406919292

Unknown billing: task 283 Late Selective Patch had an APIConnectionError at agent round 5 followed by the preconfigured retry. The failed request has no usage or fee data. Recorded aggregate tokens and complete USD therefore remain null for that agent execution; successful-response usage supplies a known lower bound only. The total above is not an exact invoice.

Update token/API amounts were checked against raw updater response usage; update elapsed time comes from stored adaptation timer records. Agent and judge costs were recomputed from their response usage, including failed-request uncertainty. Local timings are not independent wall-clock measurements. Update costs are not multiplied by the five judge ratings.

Historical development, diagnostics and confirmation runs are separate costs. Six retention diagnostics cost a recorded $0.0527461802; the main-cohort known total excludes them. Failed confirmation rounds and stopped connectivity diagnostics retain their own logs and incomplete bills in the historical archive set.

Evidence: evidence/recomputed/metrics.json costs and unknown_billing; adaptation/updater-audit.json; raw trace request_retry event; history archive manifest.

7. Failure evidence: under-update and confounds

We classify a stated claim as **confirmed** (direct mechanism evidence), **suggestive** (plausible but confounded), **inconclusive** (insufficient evidence), or **contradicted** (inconsistent with stored knowledge/calls). Confirmation of a local failure does not establish a general strategy effect.

U1 - Confirmed stale No Update call failure

Task 133 Late: Early validate_syntax requires expression only; Late also requires allow_variables. No Update keeps the Early schema. Four initial validation calls omit allow_variables and are rejected; later calls add the parameter after reading errors. The agent completes with TF 9.8/PE 7.2 after 38 calls. The stale-schema-to-rejection chain is direct, but it is a No Update observation, not a Selective Patch under-update example.

U2 - Contradicted omission attribution for Patch

The main Selective Patch artifact contains the required field and its three validate_syntax calls explicitly pass allow_variables. It completes with TF 10/PE 7, four total calls and zero tool errors. Calling this a missed parameter update would contradict the actual records.

B1 and P1 - Different failure mechanisms

Task 133 Early Patch truncates in round 2 with TF 9/PE 10, although every Early arm has identical K0. An update-caused explanation is contradicted. Task 283 Late Patch reports review_workspace_health failing because listMemories is missing; it also has two empty-argument create_task errors, an API retry and round exhaustion. The tool error is observed directly, but its contribution to the terminal failure is not isolated. Neither should automatically be called under-update or over-update.

Audit scope

We screened all 317 accessible raw_trace.json files, with comparable old/updated/target metadata for 298. The 72 structural candidates comprise 31 No Update stale snapshots, 40 Full Rewrite cases with edits to metadata-unchanged tools, and one linked_notes case. Six stale-parameter mechanisms are confirmed, two cases are suggestive and 64 remain inconclusive. Nine selected case studies receive deeper semantic review. The screen does not prove absence of all semantic usage-rule omissions; 19 traces lack complete comparable metadata.

Requirement gap: no confirmed natural under-update trajectory of the original evaluated Selective Patch has been identified. We do not promote a baseline failure to satisfy that requirement.

Evidence: evidence/trajectory-inventory.csv; trajectory-screening.json; failure-cases.json; case-details/U1.json, U2.json, B1.json, P1.json. Each case includes exact old/updated/target fields and calls.

8. Failure evidence: over-update and negative results

O1 - Suggestive historical Full Rewrite regression

In the older task 235 Late cohort, all 28 chart records are unchanged across environments. Full Rewrite nevertheless regenerates every description. Its trajectory makes five map calls and omits map-unit metadata, with TF 8.8/PE 4.2 versus 10/10 for the other arms. However, the rewritten description explicitly mentions dataValueUnit. The omission therefore cannot be explained as guidance simply disappearing. The regression is not itself proof that rewriting caused damage, and this is Full Rewrite rather than Selective Patch.

O3 - Inconclusive causal attribution after negative diagnostics

Six existing controlled description-rollback executions, three per arm, all complete with TF/PE 10 and three tool calls. They do not reproduce the historical damage. We retain this negative result and do not repeat it until a desired effect appears. It does not establish that harmful rewrites are impossible.

O2 - Suggestive scope expansion in linked_notes

A historical research variant adds this rule to metadata-unchanged math-mcp::evaluate:

validate_syntax now accepts optional detailed and allow_variables flags; evaluate's validate_only remains a separate syntax-only path, so choose based on whether evaluation is also wanted.

The task 133 Middle trajectory performs two syntax checks and a security-prompt call, then three evaluate calls. It completes with TF 10/PE 5; original Patch in that cohort uses three calls and scores 10/10. Four usage-rule records differ between these arms, so the note is not an isolated treatment. Also, evaluate is semantically related to the changed validator: metadata-unchanged does not necessarily mean unrelated. This is a suggestive research-variant case, not a confirmed failure of the main method.

Judge limitation: correct scores are not corrected answers

A fresh factual review of the six diagnostic answers finds three claiming the monthly sales sum is 24.4M: rewrite-r0, rewrite-r2 and rollback_descriptions-r0. The supplied 12 numbers sum to 25.4M. All three retain their original TF/PE 10. Errors in both arms weaken any claim of rewrite-specific factual harm and demonstrate that high judge ratings can miss answer defects.

Requirement gap: no confirmed causal over-update case for the original Selective Patch. The cases above remain labelled according to the strength and scope of their evidence.

Evidence: case-details/O1.json and O2.json contain exact descriptions/rules; diagnostic-review.json identifies all six trajectories; raw answers and five ratings remain unchanged.

9. Interpretation and limits

Supported observations

Selective Patch updates tool knowledge without model calls and prevents the specific initial missing-parameter errors seen in No Update task 133 Late. In this cohort it retains task 235 TF and has the highest mean PE. Full Rewrite leads the mean TF and ECS comparisons; both baselines have more normally completed runs. These outcomes can coexist and do not require a composite winner.

What the data do not establish

One execution per cell on three tasks is not enough to estimate robust strategy-level uncertainty, frequency of rare failures or generalization to the full benchmark. Temperature0 does not make provider execution deterministic. Five judge ratings characterize one saved trajectory, not run-to-run agent variance. No inferential significance claim is made for the27-run main cohort.

TF is near the scale ceiling, and incomplete trajectories can receive high TF. Early differences under identical K0 affect cross-version scores. The same model family acts as agent and judge. Rubric interpretation of distinct security checks and redundant calls is imperfect; manual factual findings are reported separately, never silently substituted for ratings.

The subset lacks required tool removal/renaming, uses optional tool additions, and is not an unseen-family holdout. The runner uses native stdio and common local adaptation plumbing, rather than an exact recreation of every paper infrastructure choice. Full Rewrite and Patch differ in both update scope and generation mechanism. Public MCP package defects are kept common across arms.

Reuse of prior work

The completed 108-trajectory research confirmation contains 81 executions of the original three methods with three repeats each. It can serve as a separate robustness appendix, alongside its fourth candidate and negative promotion result. Its frozen Full Rewrite artifacts and shared-build protocol differ from the main cohort, so results are not pooled. Sixty research development trajectories, old A/B cohorts, incomplete confirmations, generation rejections and negative diagnostics remain available as history.

The historical PE+0.5 and statistical-promotion thresholds are not course completion requirements. Netlify/task 223 is an optional custom deployment demonstration; it is not required to complete the fixed 27-cell experiment and is not run here. VPN involvement is possible in past connection failures, but not established as their cause.

Stopping rule

Stop strategy optimization. Finish evidence-linked reporting and review. If strict failure-case evidence remains insufficient, state the gap and seek approval for a precise small diagnostic or clarification of acceptable case scope. Do not run further 81/108-cell matrices, select a better-looking cohort or manufacture natural failures.

10. Reproducibility and delivery

The package includes this report, a link to the editable shared presentation and speaker notes, independent scripts, machine-readable calculations, frozen source/configuration archives, the original 27-run raw archive, an indexed trajectory corpus and historical experiment archives. SHA-256 manifests identify exact files. No API credentials are included. The package is prepared locally; no external submission or deployment has occurred.

Offline recalculation - no credentials or dependencies

```
python3 reproduce/recalculate.py
# Optional explicit paths:
python3 reproduce/recalculate.py --archive evidence/main-raw.tar.gz \
--output evidence/recomputed-check
```

The script reads the archive directly without extraction, verifies its identity and terminal files, recomputes all 135 ratings and 27 cells, checks execution status from trace endings, compares settings and static rubric, and calculates costs including unknowns. It does not import the experiment implementation or make network requests. CSV and JSON outputs are regenerated independently.

Optional future execution - authorization required

```
mkdir -p frozen-source
tar -xzf evidence/source-main-12d693c.tar.gz -C frozen-source
cd frozen-source
uv sync --frozen
# Install pinned Node and released MCP environments per RUNBOOK.html.
# Configure credentials locally; do not put them in the package.
./experiment prepare --suite experiments/benchmark-v1.3.json \
--repeats 1 --output results/raw/approved-new-run
./experiment run --output results/raw/approved-new-run
```

These commands are documentation, not an instruction to rerun the completed experiment. A fresh execution may not reproduce identical numerical scores because the provider is stochastic. Reproduction of archived arithmetic is exact; fresh behavioral reproduction is a separate, billable activity requiring approval.

Provenance

Main source commit: 12d693c. Benchmark commit: 888c779a0fe735f612ca62bdd1fdd7c9e6bf8701. Dataset revision: 162e3ffbea1717ac8a20fe9b26465e29117229e4. Runtime recorded as macOS arm64, Python3.12.12, Node 24.13.0; Python dependencies are locked in uv.lock. Full released environment archives require a separate large download; the raw arithmetic replay does not.

```
Main archive SHA-256:
e95e15694abe9a12e37c813f81e6e0fb0b5ccd00c2003f2a25a36a092c8001c9
```

References

[1] PE 6203 Group Project 4: Tool Harness Adaptation under Environment Change, course brief, §§3-4. [2] Liu et al., MCPEvol-Bench: Benchmarking LLM Agent Performance Across Dynamic Evolutions of MCP Servers, arXiv: 2607.14642v1, 2026, §6.1 and AppendixD.1. https://arxiv.org/html/2607.14642v1 . [3] Frozen benchmark release and course implementation identified above. [4] Local immutable experiment and diagnostic records indexed in evidence/.

Appendix A. All 27 scored cells

Task	Stage	Method	TF	PE	Execution status
133	early	No Update	9.8	8	output_truncated
133	early	Full Rewrite	10	8	completed
133	early	Selective Patch	9	10	output_truncated
133	middle	No Update	10	7.8	completed
133	middle	Full Rewrite	9.8	5.6	completed
133	middle	Selective Patch	10	8	completed
133	late	No Update	9.8	7.2	completed
133	late	Full Rewrite	10	5.8	completed
133	late	Selective Patch	10	7	completed
235	early	No Update	10	10	completed
235	early	Full Rewrite	10	10	completed
235	early	Selective Patch	10	10	completed
235	middle	No Update	10	4.8	completed
235	middle	Full Rewrite	10	10	completed
235	middle	Selective Patch	10	10	completed
235	late	No Update	10	10	completed
235	late	Full Rewrite	10	10	completed
235	late	Selective Patch	10	10	completed
283	early	No Update	10	8	completed
283	early	Full Rewrite	10	9	max_rounds
283	early	Selective Patch	10	9	completed
283	middle	No Update	10	7.4	completed
283	middle	Full Rewrite	10	7.8	completed
283	middle	Selective Patch	10	9.4	max_rounds
283	late	No Update	10	7.6	max_rounds
283	late	Full Rewrite	10	9	max_rounds
283	late	Selective Patch	10	8.6	max_rounds

Every row averages five valid ratings. Status is retained independently of the scores. Raw rating arrays, exact fractions, tool-call counts and per-cell usage are available in evidence/recomputed/metrics.json and cells.csv.

Appendix B. Completion checklist

Required deliverable	State	Evidence / outstanding work
Selective method and two baselines	Ready	Frozen source, configs and27 real trajectories.
TF/PE/ECS and execution status	Verified	Independent raw-archive recalculation; both ECS conventions.
Recovery and retention	Verified, limited scope	Two changed tasks and one unchanged task; no broad claim.
Update cost	Verified with uncertainty	Tokens/time/API separated; one main failed-request charge unknown.
Under-update failure trajectory	Strict gap remains	Confirmed No Update stale calls; not a confirmed natural Selective miss.
Over-update failure trajectory	Strict gap remains	Two suggestive cases; negative rollback diagnostics retained.
Report and reproducibility	Prepared for review	This report, scripts, logs, source and manifests.
10-minute presentation + Q&A	Prepared for review	Editable deck, timed speaker notes and Q&A; group rehearsal outstanding.
Final submission	Not performed	Team authorship/contribution review and submission destination confirmation remain.

The requirements are not declared fully satisfied while strict case evidence is missing. A transparent report of an unobserved failure class is scientifically preferable to false attribution, but whether it satisfies the course deliverable needs instructor/TA judgment. Any proposed controlled fault injection must be distinguished from a naturally occurring benchmark failure.

The next permissible work is offline review and group preparation. Additional model calls, diagnostics, task/repeat expansion and Netlify deployment remain disallowed without explicit approval.